INCH-POUND

MIL-P-70525 (AR)
29 OCTOBER 1990

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
MOTOR BODY BONDING ASSEMBLY FOR

This Specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication and packaging of the motor body, nozzle and motor body bonding assembly for 105MM, HE, RA, Projectile XM913/XM927 (see 6.11).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a parts of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Original Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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MILITARY

MIL-P-116	- Preservation-packaging, Methods of
MIL-P-117	- Bags, Sleeves and Tubing-Interior Packaging
MIL-A-2550	- Ammunition, General Specification For
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-R-13562	- Rotating Bands, Welded Overlay, Ferrous and Nonferrous, for Projectiles
MIL-A-48078	- Ammunition, Standard Quality Assurance provisions, General Specification For

STANDARDS

FEDERAL

FEDERAL TEST METHOD

STD No. 141	- Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing
STD No. 151	- Metal, Test Method

MILITARY

MIL-STD-410	- Nondestructive Test Personnel Qualification and Certification
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg 4D , 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the specification.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9381133	- Body, Motor
9381134	- Insert, Nozzle
9381136	- Motor Body Bonding Assembly

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INSPECTION EQUIPMENT DRAWINGS

9280383	- Envelope Drawing for Magnetic Particle Inspection, Motor Body, 105MM, HE, RA, XM913

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity for bids or request for approval shall apply.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8	- Tension Testing of Metallic Materials
E10	- Brinnell Hardness of Metallic Materials, Test Method for
E103	- Rapid Indentation Hardness Testing of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on drawing (dwg.) 9391136, associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Rotating band.

3.3.1 Minimum base metal penetration. There shall be penetration of the base metal at the weld interface in the form of arc cratering or intergranular penetrants such that a proper metallic bond is achieved between the overlay and base metal.

3.3.2 Maximum base metal penetration. there shall be no evidence of arc cratering, except for where the starting weld pass is made, when the base metal is machined to 4.000 inches diameter minimum. There shall be no evidence of arc cratering at any location on the base metal when it is machined to 3.900 inches diameter minimum.

3.3.3 Cracks.

3.3.3.1 Base metal, 3.960 inch diameter minimum. There shall be evidence of not more than four (4) cracks or intergranular penetrants at the 3.960 inch diameter except for where the starting weld pass is made. No single indication shall be greater than 1/4 inch in length and only those indications perpendicular to the longitudinal axis are permitted.

3.3.3.2 Base metal, 3.900 inch diameter minimum. There shall be no evidence of cracks or intergranular penetrants at the 3.900 inch diameter.

3.3.4 Overlay iron content. The iron content of the overlay shall not exceed 1.5 percent.

3.3.4.1 Overlay zinc content. The zinc content of the overlay shall not be greater than 12 percent.

3.3.5 Finished machined welded band.

3.3.5.1 Cracks. There shall be no cracks on the surface of the finish machined welded band.

3.3.5.2 Seams. There shall be no seams, extending across more than 1/2 inch machined welded band.

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3.3.5.3 Surface voids. Voids at the surface of the finish machined welded band shall comply with applicable requirements of MIL-R-12562.

3.3.5.4 Iron deposits. There shall be no iron deposits in excess of 1/8 inch in diameter.

3.4 Metal defects. There shall be no cracks, seams, laminations, or laps in the machined motor body as evidenced by magnetic particle inspection, except that indications no greater than 1/8 inch long are permitted in the undercut aft of the rotating band.

Indications no greater than 0.10 inch in length are permitted .10 inch forward of the copper/steel interface, forward of the band, as evidenced by magnetic particle inspection. Indications will be relatively straight and longitudinal. Indications curving or tending to curve from longitudinal to circumferential will not be permitted.

3.5 Base metal cratering.

3.5.1 Copper weld overlay seat. The frequency and distribution of cratering shall comply with the visual standards submitted in accordance with 3.5.3.

3.5.2 Copper welded overlay seat (subsurface). There shall be no cratering, except for the starting weld zone, at the 3.840 inch diameter below the copper welded overlay seat. There shall be no cratering at the 3.770 inch diameter.

3.5.3 Visual standards. Visual standards for evaluation of cratering in accordance with 3.5.1 shall be established at the time of first article sample submission.

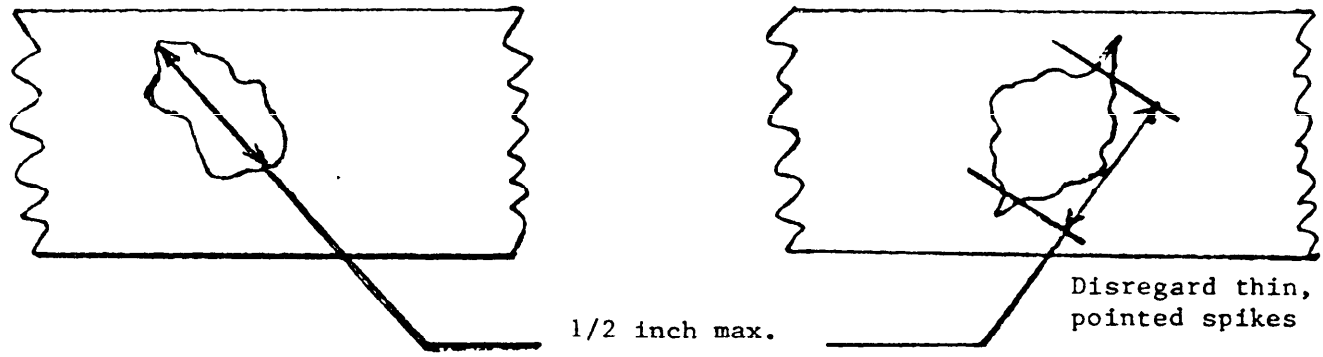
3.6 Ultrasonic inspection. Ultrasonic C-Scan inspection shall be used to inspect for quality of bond of the rotating band after heat treat.

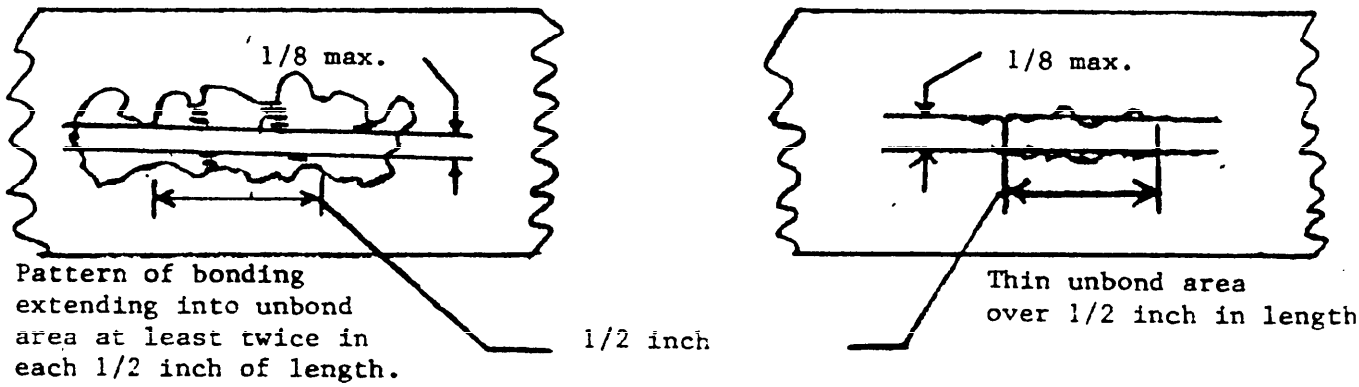
3.6.1 Linear requirement. Linear measurements shall be made with a scale having 1/16 inch graduations. The maximum linear dimensions of any unbonded area shall be 1/2 inch maximum. Thin, pointed spikes shall be disregarded. Additional definition of this requirement is made in figure 1.

3.6.1.1 Edge unbond areas. An unbond extending along either edge of the band shall have a maximum width (from the edge) of 1/3 inch and a maximum length of 3/4 inch. Any two such areas must be separated by 1/4 inch of bonding to the band edge.

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Acceptable Conditions





Unacceptable Conditions

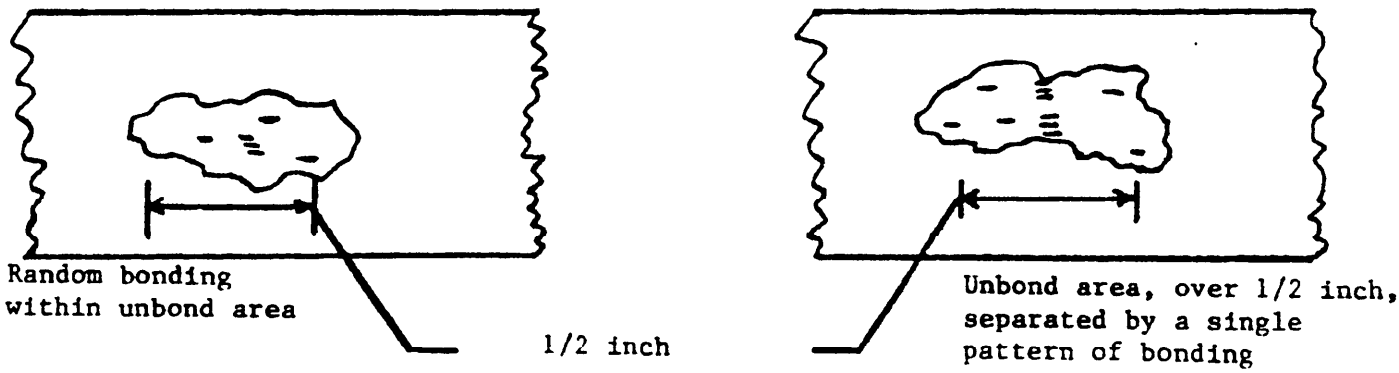
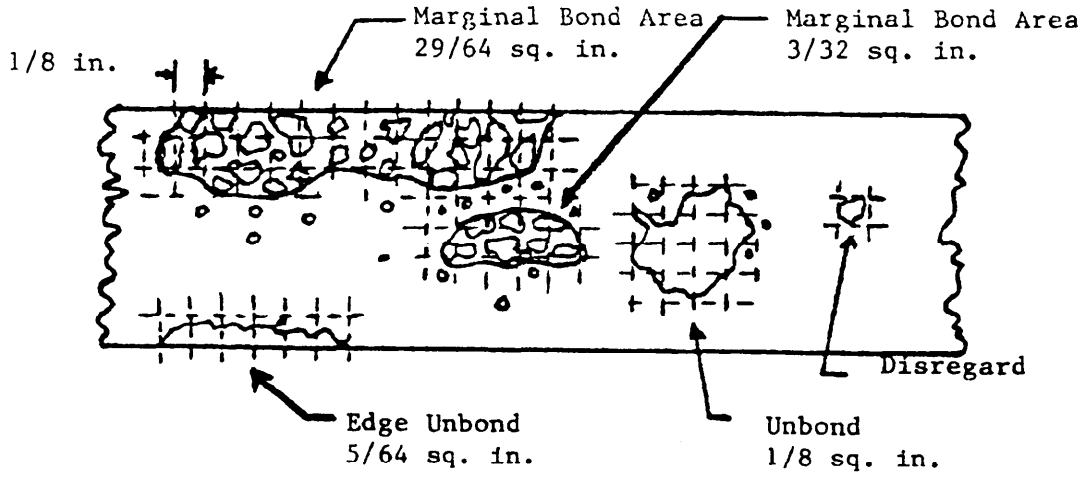


FIGURE 1. **Maximum linear dimension of unbond**

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3.6.2 Area requirements. The maximum total area of unbonds and marginal bonds shall be 1.5 square inches. A marginal bond area is any pattern of small unbonds. A clear plastic template with 1/3 inch square grids shall be used to visually sum unbonded and marginally bonded areas (see Figure 2). Unbonds of less than 1/64 square inch, that are not part of a pattern, shall be disregarded.

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Notes:

1. Total area of illustrated unbonds and marginal bonds is 3/4 square inches (sq. in.).

2. The border of a marginal bond area along the edge of the rotating band shall be the edge.

3. All grid squares within the border of a marginally bonded area shall be counted.

4. Grid squares along the border of an unbonded or marginally bonded area shall be counted if the border includes 1/2 or more of the square.

FIGURE 2. Summing unbonds and marginal bonds.

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3.6.3 Rotating band prior to ultrasonic inspection. the rotating band diameter prior to ultrasonic inspection shall be of a contractor specified diameter with tolerance of .015 inch total with a surface finish of 125 RMS or better for its full width. This requirement is to assure that the contractor will inspect to the same diameter each time. The diameter in this case shall be prior to final machining.

3.7 Proving ground test. The motor body bonding assembly shall not collapse, separate or break up in the gun bore or in flight. Testing shall be as specified in 4.5.6.

3.8 Nozzle insert.

3.8.1 Compression test. The nozzle insert shall withstand a compression load of 5000 pounds, minimum without cracking or breaking.

3.8.2 X-ray. (See Figure 1).

Zone A - There shall be no voids greater than 0.030 inch in diameter.

Zone B - There shall be no voids greater than 0.060 inch in diameter. There shall be no voids greater than 0.030 diameter which are closer than 0.060 inch apart.

3.9 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI - B46.1 and Pangborn Shot and Grit Catalog shall be used as a basis for surface roughness determination.

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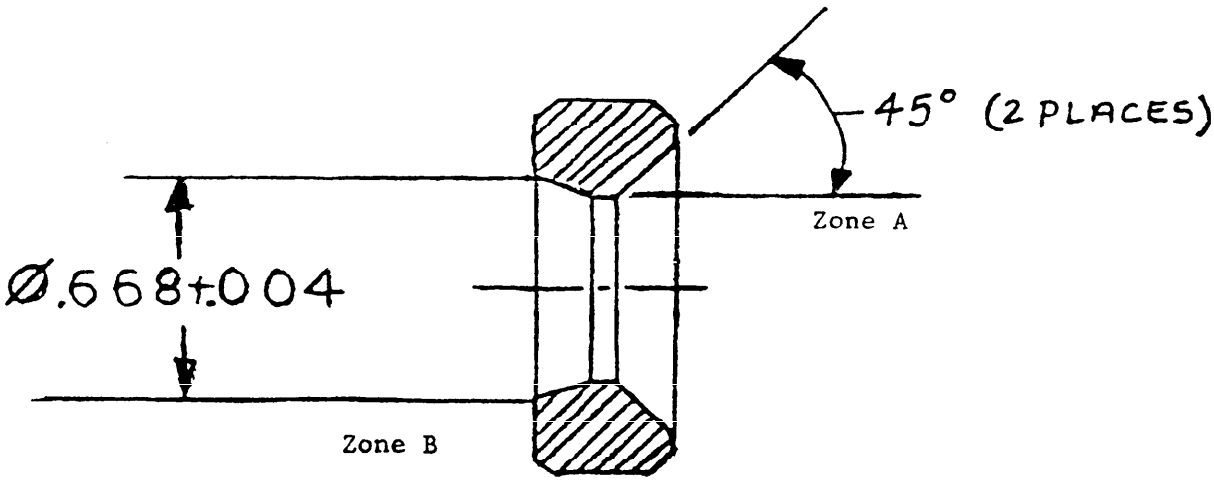


FIGURE 3. X-ray of nozzle.

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3.10 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use.

3.12 Workmanship. The requirements for workmanship are as shown on the applicable drawing in the applicable specification and the following:

3.12.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced therein. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.12.2 Painting. Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.12.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.12.4 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.12.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.12.6 Surface finish. The requirements for surface finish are as detailed on the applicable drawings. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

3.12.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor or the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements. However, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article.

4.3.1 Inspections and tests (non-ballistic).

4.3.1.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Finished Machined Body Bonding Assemblies, Painted	9381136	20
Finished Machined Motor Bodies, Unpainted and Unknurled	9381133	15
Nozzle Insert	9381134	20
Unbanded Motor Bodies	9381133	5

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4.3.1.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. Visual standards for examination of band seat after chemical removal of rotating band shall be prepared from the first article sample for acceptable visible intergranular penetration and proper distribution of arc cratering. Unless otherwise directed by the Contracting Officer, the assemblies used for ballistic testing shall not be submitted prior to acceptance of the "Inspection and tests (non ballistic)" samples submitted to the Government.

4.3.2 Ballistic tests.

4.3.2.1 Submission. Twenty complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1.2).

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Critical defect process control. The contractor shall identify all elements of this process which may cause the critical defects defined in 4.4.2. All identified elements shall be controlled. Records shall be kept of variable data when available or of the malfunction of any identified element if its function is not compatible with the variable measurement. Data recordings shall be made at maximum intervals of four hours during all shifts when product is being processed. When a critical defect is detected, the data records shall be reviewed to determine if one or more process elements can be identified as its cause. If such identification is made for a process element where variable data is recorded, the data shall be used to bring that process element into statistical control as defined in 4.4.2b. When such identification is made for any other element, suitable correction shall be applied. Deliverable data required by this specification is cited in the contract or purchase order on the Contract Data Requirements List (DD Form 1423) (See 6.4).

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER See below
	Motor Body Bonding Assembly		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Finished Machined Body Bonding Assemblies, Painted (Dwg. 9381136)			
	Examination for defects	20; 100%	3.2	4.4.2.3/4.4.2.5
	Finished Machined Motor Bodies, Unpainted and Unknurled			
	Examination for defects	15; 100%	3.2	4.4.2.2
	Magnetic particle test	15; 100%	3.4	4.5.2
	Properties of rotating band weld	2; 100%	3.3	4.4.3.2/4.5.2
	Mechanical properties of motor body	2; 100%	3.2	4.4.3.2/4.5.5
	Nozzle Inserts (Dwg. 9381134)			
	Examination for defects	20; 100%	3.2	4.4.2.4
	Compression test	5; 100%	3.6.1	4.5.3.1
	Radiographic inspection	20; 100%	3.6.2	4.5.3.2
	Unbanded Motors Bodies (Dwg. 9381133)			
	Examination for defects	5; 100%	3.2	4.4.2.1

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following:

4.4.1.1 Heat treatment inspection lot. Heat treatment inspection lots shall not exceed one mill heat of steel. Whenever a break in heat treating of eight (8) hours or more occurs or a furnace is shut down for any length of time, the heat treatment inspection lot will be terminated and a new lot shall start. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into larger heat treatment inspection lots after qualifying with individual batches provided qualifying procedures are submitted to and approved by the Government and provided all the other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality Conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum, the conformance criteria specified herein. Attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels cited in the Classification of Characteristics paragraphs.

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TABLE II. 0 - 1 Attributes Plans

Lot Size	Inspection Levels I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70525 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Motor body, Prior to Painting	1 OF 1	9381133
			NEXT HIGHER ASSEMBLY: 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of rotating band groove	Level III	3.2	Gage
Minor				
201	Width of band groove	Level V	3.2	Gage
202	Surface finish of band groove improper	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 4	DRAWING NUMBER 9381133
4.4.2.2	Motor Body, Prior to Knurl and Paint		NEXT HIGHER ASSEMBLY 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Magnetic particle test	100%	3.4	4.5.2
Special				
1	Magnetic particle test stamp missing	100%	3.2	Visual
Major				
101	Length	Level III	3.2	Gage
102	Inside diameter of cavity	Level III	3.2	Gage
103	Runout of inside diameter of cavity	Level III	3.2	Gage
104	Diameter of inner thread undercut (if applicable)	Level III	3.2	Gage
105	Diameter of outer thread undercut	level III	3.2	Gage
106	Runout of thread undercut	Level III	3.2	Gage
107	Width of inner thread undercut	Level III	3.2	Gage
108	Depth of thread (including undercuts if applicable)	Level III	3.2	Gage
109	Depth of outer thread undercut	Level III	3.2	Gage
110	Thread pitch diameter (max.)	Level III	3.2	Gage
111	Thread minor diameter (max.)	Level III	3.2	Gage
112	Runout of pitch diameter (at 2 locations) including seating of gage	Level III	3.2	Gage
113	Depth of cavity	Level III	3.2	Gage

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 4	DRAWING NUMBER 9381133
4.4.2.2	Motor Body, Prior to Knurl and Paint		NEXT HIGHER ASSEMBLY 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
114	Location of forward cavity basic diameter	Level III	3.2	Gage
115	Runout of forward cavity basic diameter	Level III	3.2	Gage
116	Length of alternate full form thread, min.	Level III	3.2	Gage
117	Depth to end of alternate thread mount, max.	Level III	3.2	Gage
118	Location of aft cavity basic dia.	Level III	3.2	Gage
119	Runout of aft cavity basic dia.	Level III	3.2	Gage
120	Depth of nozzle cavity	Level III	3.2	Gage
121	Diameter of nozzle cavity	Level III	3.2	Gage
122	Runout of nozzle cavity wall	Level III	3.2	Gage
123	Location of forward delay seat basic diameter	Level III	3.2	Gage
124	Runout of forward delay seat basic diameter	Level III	3.2	Gage
125	Location of aft delay seat basic diameter	Level III	3.2	Gage
126	Runout of aft delay seat basic dia.	Level III	3.2	Gage
127	Diameter of bottom of delay seat bore relief	Level III	3.2	Gage
128	Depth of moat measured from aft boattail and aft delay hub (120 apart, 3 places)	Level III	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Motor Body, Prior to Knurl and Paint	3 OF 4	9381133
			NEXT HIGHER ASSEMBLY 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
129	Location of forward boattail basic diameter	Level III	3.2	Gage
130	Runout of forward boattail basic diameter	Level III	3.2	Gage
131	Location of aft boattail basic dia.	Level III	3.2	Gage
132	Runout of aft boattail basic dia.	Level III	3.2	Gage
133	Diameter of delay hub	Level III	3.2	Gage
134	Radius at cavity base	Level III	3.2	Gage
135	Rotating band diameter	Level III	3.2	Gage
136	Runout of rotating band	Level III	3.2	Gage
137	Width of rotating band (incl. bevel)	Level III	3.2	Gage
138	Width of rotating band major dia.	Level III	3.2	Gage
139	Angle of rotating band bevel	Level III	3.2	Gage
140	Diameter of groove, aft of band	Level III	3.2	Gage
141	Outside diameter (forward and aft of band)	Level III	3.2	Gage
142	Mechanical properties of body	Level III	3.2	4.4.3.1/4.5.6
143	Properties of rotating band weld	Level III	3.2	4.4.3.2/4.5.2
144	Location of rotating band	Level III	3.2	Gage
145	Surface finish of outer thread undercut	Level III	3.2	Visual/ Comparison Blocks
146	Presence of surface voids, seams, iron deposits or cracks in rotating band	Level III	3.2	Visual/Probe/Scale

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70525 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.2	Motor Body, Prior to Knurl and Paint	SHEET 4 OF 4	9381133
			NEXT HIGHER ASSEMBLY 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Diameter of delay seal bore	Level V	3.2	Gage
202	Depth of delay seal bore shoulder	Level V	3.2	Gage
203	Inside diameter of boattail flange	Level V	3.2	Gage
204	Width of groove, aft of band	Level V	3.2	Gage
205	Any feature missing	Level V	3.2	Visual
206	Thread pitch diameter (min.)	Level V	3.2	Gage
207	Thread minor diameter (min.)	Level V	3.2	Gage
208	Surface finish of delay seat	Level V	3.2	Visual/ Comparison Blocks
209	Dimension controlling depth of delay seal relief grooves, max. (2 places)	Level V	3.2	Gage
210	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70525 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Motor Body, After Paint	1 OF 1	9381133
			NEXT HIGHER ASSEMBLY: 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of motor body (max)	100%	3.2	Gage
102	Salt spray test	4.4.3.5.1	3.2	4.5.4.1
103	protective coating thickness	4.4.3.5	3.2	4.5.4.2
104	Knurl missing, incomplete, inadequate or contaminated by paint or foreign material	100%	3.10	Visual
Minor				
201	Protective finish incomplete or damaged, exposing base metal	Level V	3.2	Visual
202	Marking missing, misleading, inadequate, or illegible	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

AMSC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70525 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Nozzle Insert		9381134
			NEXT HIGHER ASSEMBLY 9381136

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Radiographic inspection	100%	3.6.2	4.5.3.2
102	Grain flow	4.4.3.4.2	3.2	4.4.3.4.2
103	Compression test	Level III	3.6.1	4.4.3.4.1/4.5.3.1
104	Outside diameter	Level III	3.2	Gage
105	Diameter at throat divergence (2 places)	Level III	3.2	Gage
106	Chamfer missing or improper	Level III	3.2	Visual
107	Surface finish of inside profile improper	Level III	3.2	Visual/ Comparison Blocks
108	Throat diameter	Level III	3.2	Gage
109	Perpendicularity of end surface (2 places)	Level III	3.2	Gage
110	Ink marking forward side missing or misapplied	Level III	3.2	Visual
Minor				
201	Runout of throat diameter with outside diameter	Level V	3.2	Gage
202	Total length	Level V	3.2	Gage
203	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70525 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9381136
4.4.2.5	Motor Body Bonding Assembly		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Nozzle insert not properly bonded	100%	3.2	Manual
102	Nozzle insert not flush or below	Level III	3.2	Gage
103	Orientation of nozzle insert incorrect (white side)	100%	3.2	Visual
104	Excessive adhesive in motor cavity	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Mechanical properties of motor body.

4.4.3.1.1 Hardness. Each banded motor body from a heat treated lot shall be identified and subjected to a Brinell hardness test on surface indicated. Three hardness readings shall be taken approximately 120 degrees apart. The three hardness readings thus obtained shall be averaged and used as the basis for selecting the tension test samples of paragraph 4.4.3.1.2.

4.4.3.1.2 Tension test (see 4.4.3.1.1). Two banded motor bodies from the same heat treat lot shall be selected for this test (one motor body from the highest hardness reading and one motor body from the lowest hardness reading). From each of these motor bodies, three longitudinal specimens shall be tested to provide yield strength and elongation data. One specimen shall be taken from the side wall near the boattail. The second specimen shall be taken with the gage length of the specimen directly beneath the rotating band. The third specimen shall be taken from the side wall of the threaded cavity near the motor body thread. Testing shall be conducted in accordance with ASTM, Method E8 with the largest appropriate substandard size round specimen.

4.4.3.1.2.1 Retest. If the high hardness motor body fails the yield strength requirement, the entire heat treatment inspection lot shall be rejected. In the event the lowest hardness motor body fails to meet the yield strength requirements, higher hardness values may be selected and tested in accordance with 4.4.3.1.2 in order to establish the acceptance minimum hardness. Once this minimum hardness has been established, all motor bodies with hardness values below the new minimum shall be rejected. Failure to meet elongation regardless of hardness shall be cause for rejection of the entire inspection lot. Rejected motor bodies may be retempered or completely reheat treated. Rejected motor bodies shall be preconditioned at 800 degrees Fahrenheit (F) to 1000 degrees F for 15 minutes minimum prior to austenitizing. In addition, motor bodies shall not have subsequent machining operations that leave sharp corners that could cause cracking during heat treatment. The effectiveness of the reheat treating or retemper cycle shall be determined by reheat treating or retempering the remaining portion of the motor body sample that failed with the lot and then by retesting the sample after reheat treatment or retemper. In addition, one (1) sample motor body with the highest hardness and one (1) sample motor body with highest hardness after reheat treatment or retemper shall be

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selected and tested in accordance with 4.4.3.1.2. If no portion of the originally rejected motor body remains, then two (2) sample motor bodies shall be selected and tested with the highest hardness values and two (2) sample motor bodies shall be selected and tested with the lowest hardness values. Failure of any specimen to comply with either yield strength or elongation requirements shall be cause for rejection of the reheat treated or retempered lot.

4.4.3.2 Rotating band. A sample shall be taken, for chemical analysis and minimum penetration, from each overlay band machine, from each eight (8) hour shift or fraction of a shift from machines in use. The randomly selected motor bodies to be used as samples, and all bodies they represent, shall be identified for each machine and shift until all tests and examinations of 4.4.3.2 are complete. The bodies selected are not required to be heat treated.

4.4.3.2.1 Chemical analysis. The iron and zinc content of the copper shall be determined by chemical analysis of chips machined from the copper overlay. The chips for the sample shall consist of copper overlay between 4.09-.03 inches diameter. All of the chips shall be collected and crushed, with a sample portion of 20 ± 5 grams then randomly selected and dissolved. Chemical analysis shall be conducted in accordance with Federal Test Method Standard No. 151 except that the sample shall not be cleaned with a magnet. Samples having iron content of 1.0 percent max. shall be considered acceptable. Any samplie having iron content 1.0 to 1.5 percent max. shall be considered acceptable upon demonstration of compliance, for that sample's base metal, with the requirements of 3.3.2 and 3.3.3 (per 4.4.3.2.3). Any sample having iron content in excess of 1.5 percent, and that portion of production represented by that sample, shall be considered rejected. Any sample requirement, and that portion of production represented by that sample, shall be considered rejected.

4.4.3.2.2 Minimum penetration. Each sample motor body selected in accordance with 4.4.3.2 shall be inspected for minimum base metal penetration. The remanent copper overlay shall be dissolved by any method which will not damage the weld interface surface. The remanent may be machined to 4.090 inches minimum diameter prior to dissolving the remaining copper. The surface shall be visually observed for minimum penetration as follows:

a. The sample shall be considered acceptable if the quantity and distribution of arc cratering equal or exceed that of the appropriate visual standards.

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b. The sample shall be considered acceptable if the visible intergranular penetrants, under magnetic particle inspection, equal or exceed those of the appropriate magnetic particle standard.

c. The sample shall be considered acceptable if all of the following conditions are satisfied:

1. No uncratered area exceeds 1/4 square inch. Evaluate utilizing an overlay with a 1/2 x 1/2 inch square. Overlay shall be positioned with top and bottom edges of square parallel to the edges of the band seat.

2. The sum of uncratered areas does not exceed two (2) square inches. Evaluate utilizing an overlay divided into 1/8 x 1/8 inch (1/64 square inch) boxes. Overlay shall be positioned with top and bottom edges of squares parallel to the edges of the band seat. All uncratered areas which exceed 3/8 x 3/8 or 3/4 x 1/4 inch shall be totaled to the nearest 1/64 square inch. The overlay shall be of sufficient size to cover the entire band seat and shall not be moved during evaluation of surface for items "2", "3" and "4". One uncratered area, within 1 inch of the weld stop-start, may be disregarded if the width is less than 3/8 inch.

3. Uncratered areas are separated by 1/2 inch minimum. Evaluate utilizing overlay as in item "2".

4. No uncratered area is within 1/8 inch of the forward or aft edge of the band seat. Evaluate utilizing overlay as in item "2", and disregarding one area within 1 inch of the weld stop-start.

5. Each uncratered area identified in item "2", when examined by magnetic particle inspection, exhibits two (2) or more intergranular penetrants.

Any sample failing to meet a, b or c above, and that portion of production represented by that sample, shall be considered rejected.

4.4.3.2.3 Maximum penetration and cracks. Every third day's sample, from each band machine selected in accordance with 4.4.3.2 shall be examined for maximum penetration and cracks. Any sample failing to meet the applicable criteria, when examined in accordance with 4.4.3.2.3.1, 4.4.3.2.3.2 and 4.4.3.2.3.3, and

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that portion of production represented by that sample, shall be considered rejected. If the third day's sample is rejected, the second day's sample may be examined for acceptance of the two day's production, from that machine which the sample represents. If the second day's sample, and production represented by that sample, is rejected, the first day's sample may be examined for acceptance of that day's production.

4.4.3.2.3.1 Maximum penetration (other than starting weld pass). The sample band seat shall be machined to a minimum diameter of 4.000 inches. The band surface shall be visually examined for presence of arc cratering, except for the area at the start of the weld pass. The starting weld pass area is defined as 1/2 inch maximum on each side of the start-stop weld center line.

4.4.3.2.3.2 Cracks. Upon completion of 4.4.3.2.3.1, the band seat shall be machined to a minimum diameter of 3.960 inches. The entire band seat area shall be examined, by magnetic particle inspection, for cracks and intergranular penetrants. The presence of indications in the starting weld pass area shall not be cause for rejection. If the entire band seat area is free of indications under magnetic particle inspection, and free of arc craters under visual examination, the sample shall be considered final accepted and examination under 4.4.3.2.3.3 is not required.

4.4.3.2.3.3 Maximum penetration and cracks (including starting weld pass). Upon completion of 4.4.3.2.3.2, and if required, the band seat shall be machined to a minimum diameter of 3.900 inches. The entire band seat area shall be visually examined for the presence of arc cratering, and examined by magnetic particle inspection for cracks and intergranular penetrants. Presence of any craters, cracks or penetrants shall be cause for rejection of the sample and all production represented by the sample.

4.4.3.3 Hydrogen embrittlement relief. The contractor shall comply with the requirements of TT-C-490. For verification, the contractor shall submit a certificate of compliance that motor bodies have been relieved of hydrogen embrittlement.

4.4.3.4 Ultrasonic inspection.

4.4.3.4.1 Rotating band diameter prior to ultrasonic inspection. The outside diameter and surface finish shall be measured (finish-visually) for conformance to requriements, specified in 3.6.3 according to the following inspection plan:

Inspect the first 120 motor bodies manufactured on a 100 percent basis. After meeting this, inspection may be reduced to a Level IV in Table II, 0-1 Attributes Plan in 4.4.2.

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4.4.3.4.2 Ultrasonic inspection. This inspection shall be performed 100 percent after copper deposition and heat treating, with the band diameter machined as specified in 3.6.3. Failure of the motor body to meet the requirements of 3.6 shall be cause for rejection of that motor body. This inspection shall be conducted prior to finish machining.

4.4.3.5 Nozzle insert.

4.4.3.5.1 Compression tests. A sample of five (5) nozzle inserts from each lot shall be compression tested. If any piece fails to meet the minimum compression load requirement, the lot shall be rejected.

4.4.3.5.2 Grain flow. A certificate of compliance shall be submitted with each lot that the grain flow of the nozzle insert is in accordance with the drawing. The contractor shall show evidence that adequate controls are being maintained in processing the nozzle insert from the billet or sheet to finished component to assure proper grain flow direction.

4.4.3.6 Protective coating. The sample size and acceptance criteria, the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements of TT-C-490 shall be cause for taking the corrective action prescribed therein.

4.4.3.6.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.6.2 Protective coating - thickness. Two samples shall be selected at random to represent each four-hour period, or less, of continuous paint production. The sample motor bodies shall be inspected for paint coating thickness at a minimum of four separate locations on each motor body. Any motor body failing to meet the applicable requirements shall be classed defective and every motor body painted since the last acceptable sample shall be rejected.

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4.4.3.7 Tool control dimensions.

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hours), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of the tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimensions) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.4.3.8 Proving ground tests.

4.4.3.8.1 First article test. Twenty (20) motor body assemblies from the first article sample shall be subjected to this test. If any of the sample assemblies fail to meet the requirements of 3.10, the first article sample shall be rejected. Range and deflection shall be recorded for information only. The test shall be conducted as specified in 4.5.9.

4.4.3.8.2 Production test. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, twenty motor body assemblies shall be selected at random from each lot for this test. If any of the sample assemblies fail to meet the requirements of 3.10, the lot shall be rejected. After three consecutive lots have complied with the acceptance criteria, ten samples shall be selected at random from each lot and tested. If any sample motor body fails to meet the requirements of 3.7, the lot shall be rejected. Range and deflection shall be recorded for information only. The test shall be conducted as specified in 4.5.9.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Inspection Method Reference column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification's requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual, manual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Magnetic particle inspection. The inspection shall be in accordance with specification MIL-I-6868 using the continuous, wet, fluorescent method. The suspension shall be applied to all surfaces of the entire motor body. When inspection is in accordance with 4.4.3.2.2 or 4.4.3.2.3, only the machined area beneath the rotating band must be covered with suspension. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 mililiters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. After draining the excess suspension, the motor body shall be inspected under ultra-violet light for the presence of defects. The adequacy of the inspection process shall be checked at the start of the daily production and at eight-hour intervals thereafter using a standard in accordance with dwg. 9280383. All plugs in the standard shall be clearly defined for a full 360 degrees both inside and out. If the inspection process is determined to be inadequate, the condition shall be corrected and the proceeding 8 hours' production shall be reinspected. The requirement of specification MIL-I-6868 to check system effectiveness by means of steel ring specimen shall be waived. Components which exhibit indications which are suspected of being shallow surface defects may have material removed to a maximum of 0.005 inches subject to the following restrictions:

a. These provisions for the removal of additional material shall not apply to the inspection beneath the rotating band on the samples selected in accordance with 4.4.3.2.

b. Involved areas shall be blended with adjacent surfaces to insure that no sharp edges, steps or gouges remain.

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c. The involved areas shall be independently checked for verification that all drawing dimensions (wall thickness, variations, etc.) are within acceptable limits.

d. The components shall be again subjected to magnetic particle inspection verification of defect removal.

4.5.2.1 Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.3 Nozzle insert.

4.5.3.1 Compressive test. The sample nozzle inserts shall be tested at a loading rate of 0.02 inch per minute. If the nozzle insert meets the minimum load requirement, the load shall be applied until the insert cracks or breaks up and the load recorded for information purposes.

4.5.3.2 Radiographic test. The nozzle inserts shall be radiographed with the radiographic beam parallel to the longitudinal axis. Radiographic inspection shall be in accordance with MIL-STD-453. The penetrameter shall be designed so as to assure a 2 percent sensitivity and the film density shall be 1.50 minimum.

4.5.3.2.1 Operator qualification. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.4 Ultrasonic inspection. Each banded motor body shall be inspected using ultrasonic C scan equipment to inspect the copper steel interface for disbonds. A standard shall be prepared from a welded motor body with a rotating band machined as required by 3.6.3. The body shall have a helical pattern of 12 equally spaced .045 + .003 inch diameter flat bottom holes drilled or burned from the inside diameter to the copper/steel interface within +/- .002 inches. A .003 inch maximum radius is required at the bottom of the holes. The pattern shall be 360 degrees with the first and last hole center lines located .188 inches from the band edges. The C scan equipment shall include a drum type chart recorder. The C scan equipment shall detect and record the first, last, and at least eight of the intermediate holes when calibrated.

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4.5.4.1 Verification and operational procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including description of equipment and standards (Contractor Design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by running a verification check on the calibration standard(s) at the start and finish of each production shift, as well as at four (4) hour intervals during production. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. A log book shall be maintained by the operator on each ultrasonic system which, as a minimum, documents the processing of standards, number of items inspected, changing of transducers, instrumentations, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

After completion and acceptance of the ultrasonic tests and the final verification test has passed, clearing the reviously inspected motor bodies, each acceptable motor body shall be marked in accordance with Dwg. 9381133. Any rejected motor bodies shall be clearly marked as such and removed from the acceptable motor bodies.

4.5.4.2 Operator qualification. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II. Periodic verification checks required may be conducted by the Level I operator, however, any equipment adjustment required as a result of a verification check shall be made by Level II qualified personnel.

4.5.5 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon.

4.5.5.1 Salt spray testing. Salt spray testing shall be accomplished on production or sectioned motor bodies that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned motor bodies are used, sectioning shall be accomplished to allow testing of all major areas ofthe motor

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body on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.5.1.

4.5.5.2 Protective coating - thickness. Protective coating thickness testing shall be accomplished using commercially available equipment.

4.5.6 Stress relief (if required). The contractor shall comply with the requirements of drawing 9381141. For verification, the contractor shall submit a certificate of compliance that the motor bodies have been stress relieved.

4.5.7 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The Hardness Tester shall use a standard 10 MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. Three hardness readings shall be taken 120 degrees apart.

4.5.8 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample motor body assemblies

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selected in accordance with 4.5.5.1 shall be inert loaded and assembled and fired at ambient temperature in M119 series cannon with muzzle brake using a propelling charge adjusted which produces an averge of 45,000 + 1,000-5,000 psi chamber pressure based on copper pressure gages. The propelling charges shall be uniformly temperature conditioned prior to the test. Pressure-time traces and differential pressure shall be obtained for information only for each round fired by means of a spindle tap in the breech of the cannon and a piezo gage at the projectile base location. Adequate photographic and instrumental coverage shall be provided to assure compliance with 3.7. Range and deflections shall be recorded when possible for informational purposes only. Recovery and examination of metal parts for any abnormal condition shall be accomplished for information only.

4.5.9 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample motor body assemblies selected in accordance with 4.4.3.7 shall be inert loaded and assembled and fired at ambient temperature in a M119 series howitzer using a PXR200 propelling charge. The propelling charge shall be uniformily temperature conditioned to 70 ± 2.5 degrees F prior to test. Photographic and instrumental coverage shall be recorded for compliance with 3.5. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes only.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use. The motor body covered by this specification is intended for use on the Projectile, 105MM, HE, RA, XM913.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, ARDEC, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Data requirements. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167. Deliverable data required by this specification is cited in the following paragraphs:

Data Requirement	Applicable DID
Test Report	OT-90-12009

(Copies of data item descriptions required by the contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.5 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.7 Submission of ballistic test data. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data shall be forwarded to: Commander, ARDEC, ATTN: AMSMC-QAR-Q and SMCAR-FSA-IM, Picatinny Arsenal, New Jersey 07806-5000.

6.8 Start zone. The starting weld zone as referenced to herein is defined as a zone 1/2 inch on each side of the starting pass centerline.

6.9 Weld overlay seat. The rotating band seat width specified on drawing 9381133.

6.10 Subject term (keyword) list.

Weld
Weld treatment
Magnetic particle inspection
Motor body
Radiographic inspection
Rotating band
Tool control
Weld penetration

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Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-AC28)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70525 (AR)	29 October 1990

3. DOCUMENT TITLE

PROJECTILE, 105MM, HE, RA, XM913/XM927 MOTOR BODY BONDING ASSEMELY FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone [703] 756-2340 AUTOVON 289-2340	

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